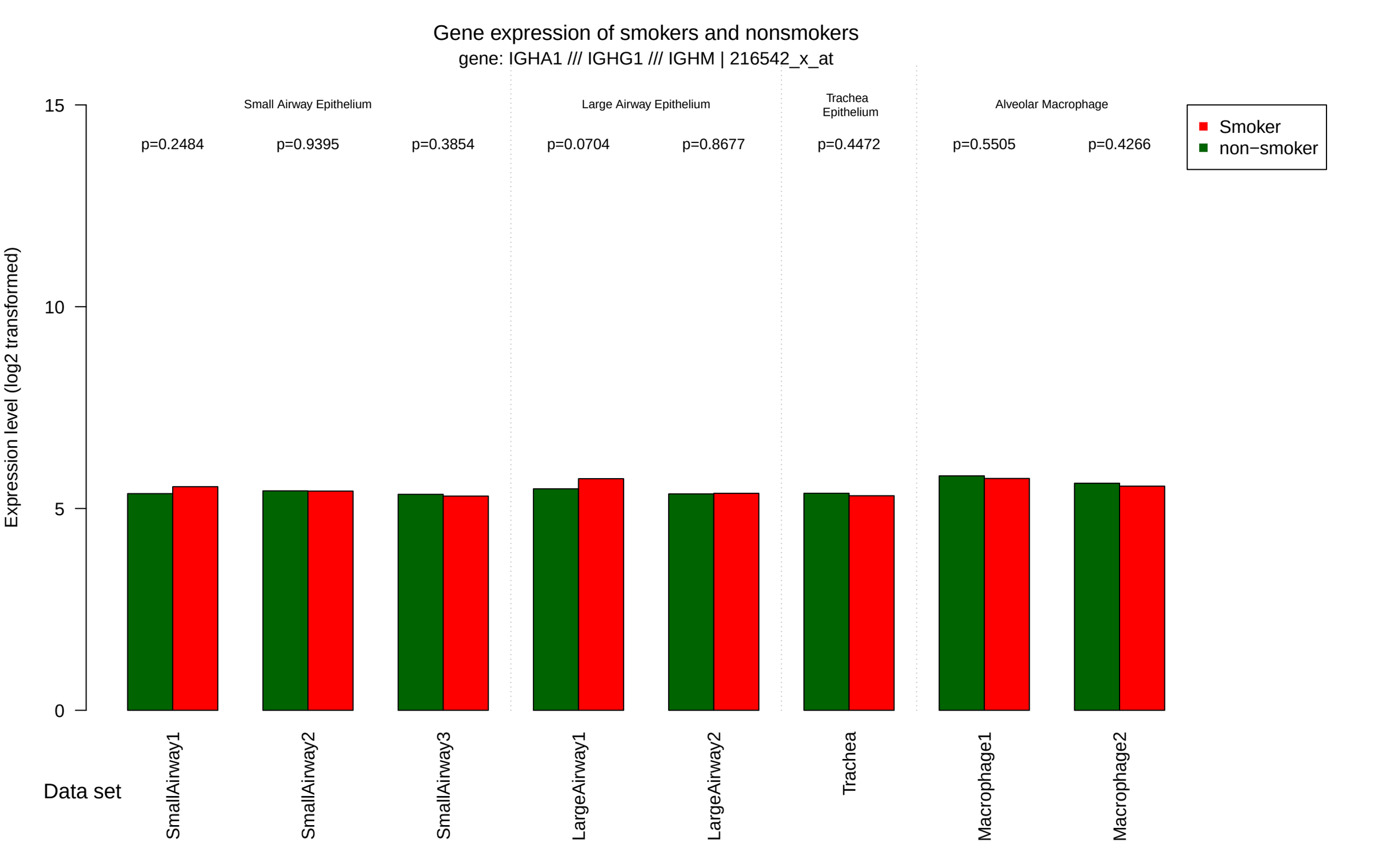
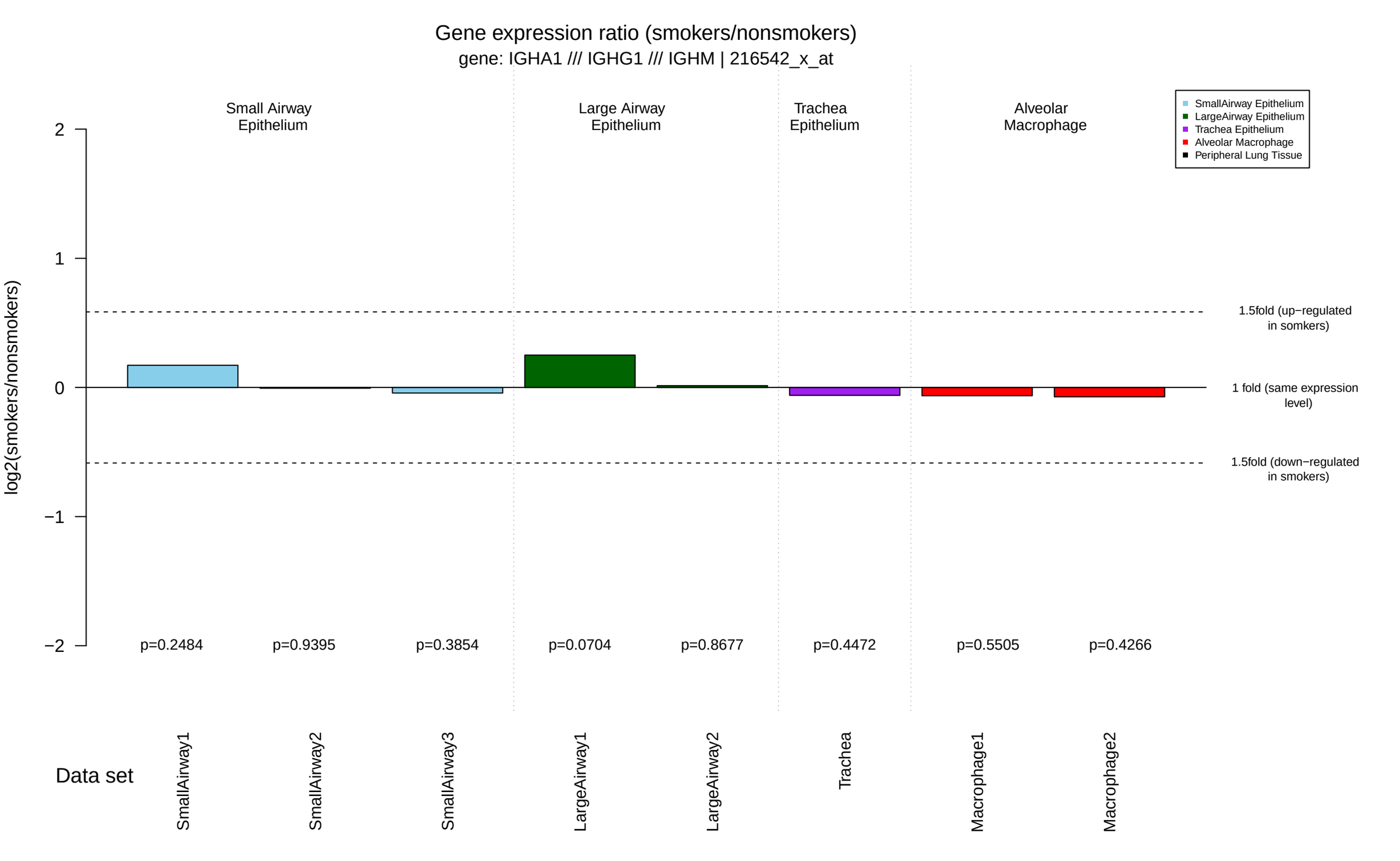
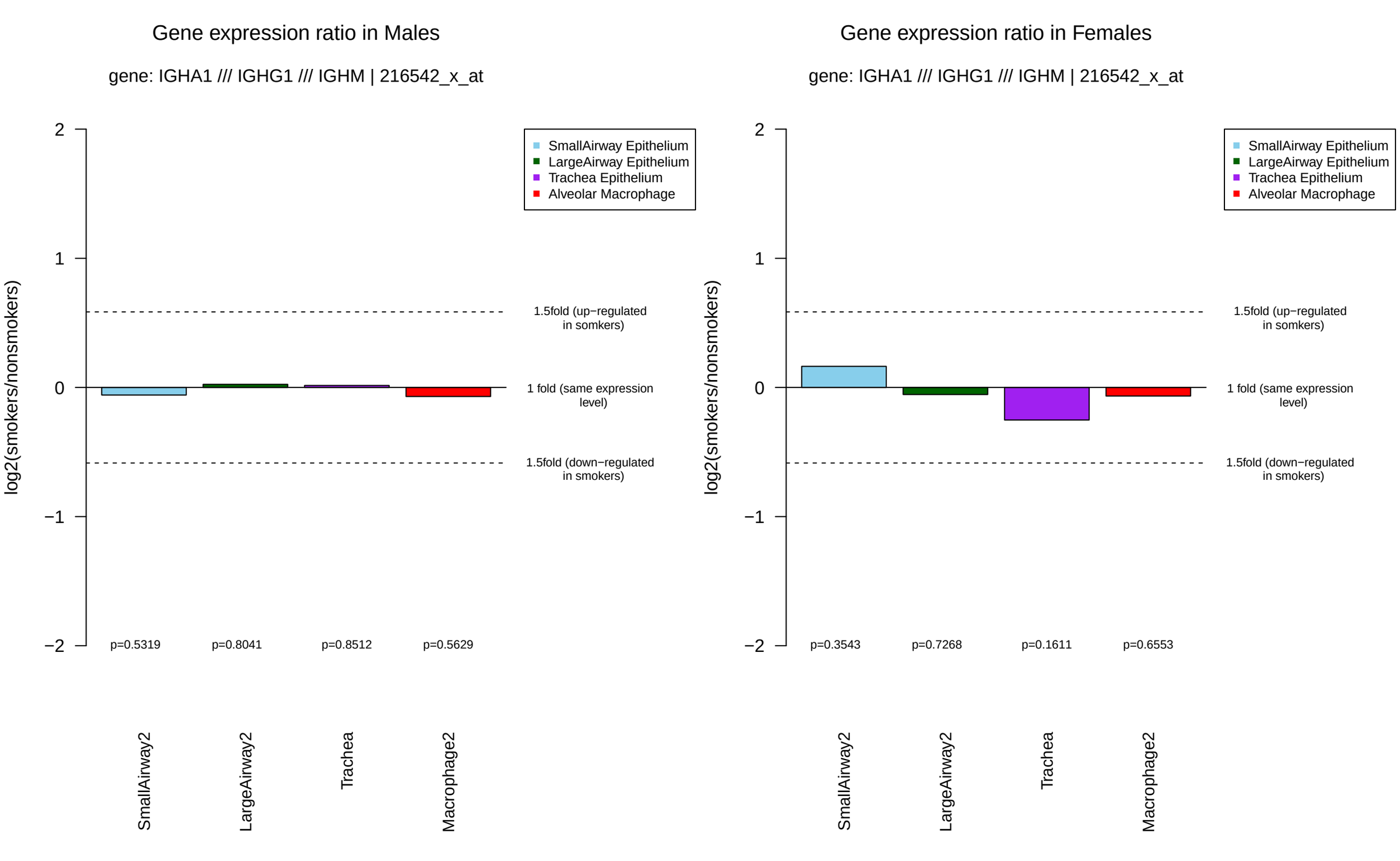
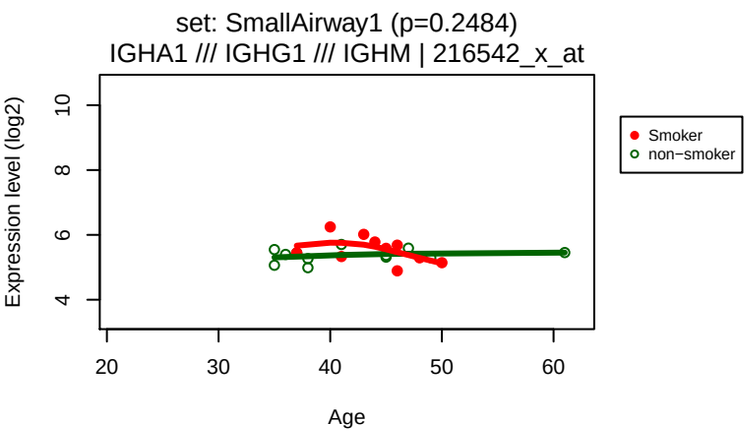
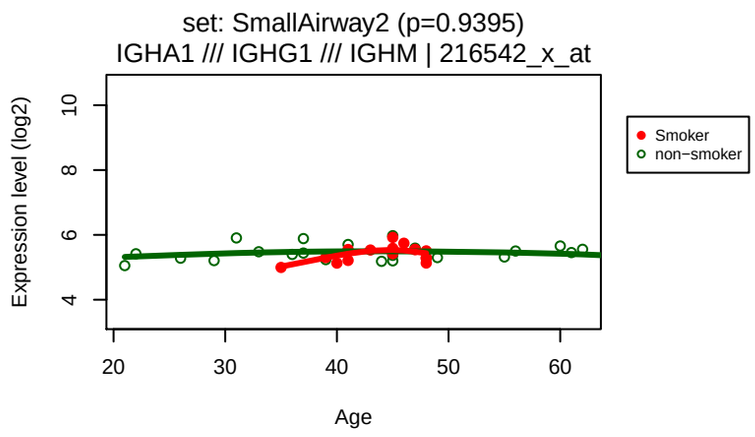
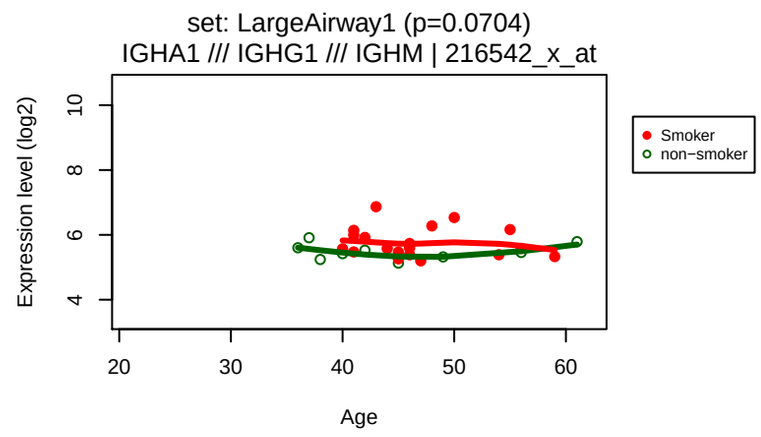

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2484
p=0.9395
p=0.3854
p=0.0704
p=0.8677
p=0.4472
p=0.5505
p=0.4266
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

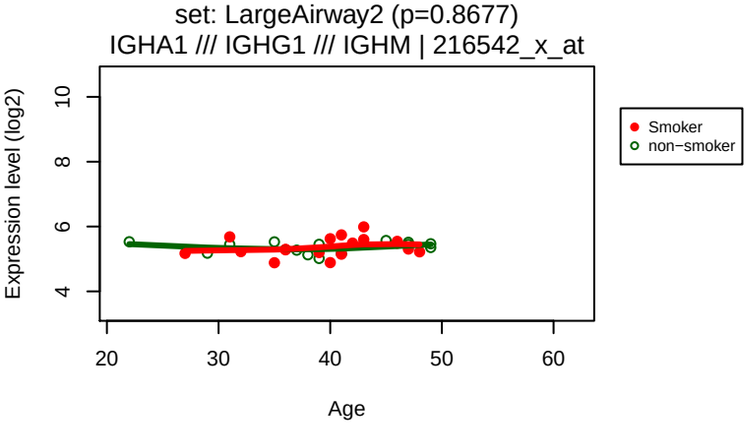

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2484
p=0.9395
p=0.3854
p=0.0704
p=0.8677
p=0.4472
p=0.5505
p=0.4266
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

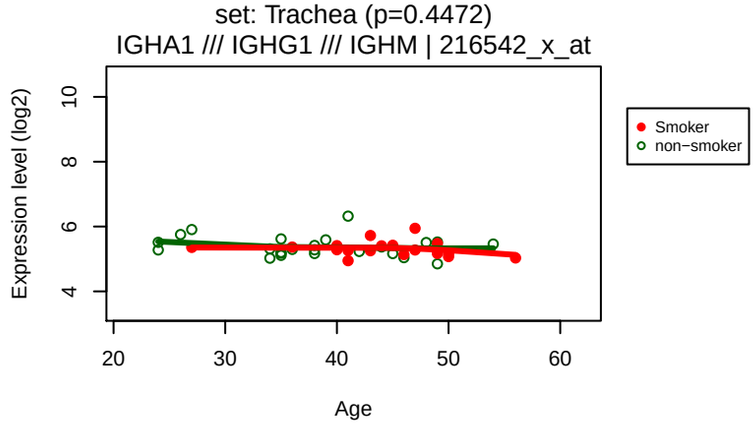

Gene expression ratio in Males
gene: IGHA1 /// IGHG1 /// IGHM | 216542_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5319
p=0.8041
p=0.8512
p=0.5629
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHA1 /// IGHG1 /// IGHM | 216542_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3543
p=0.7268
p=0.1611
p=0.6553
SmallAirway2
LargeAirway2
Trachea
Macrophage2

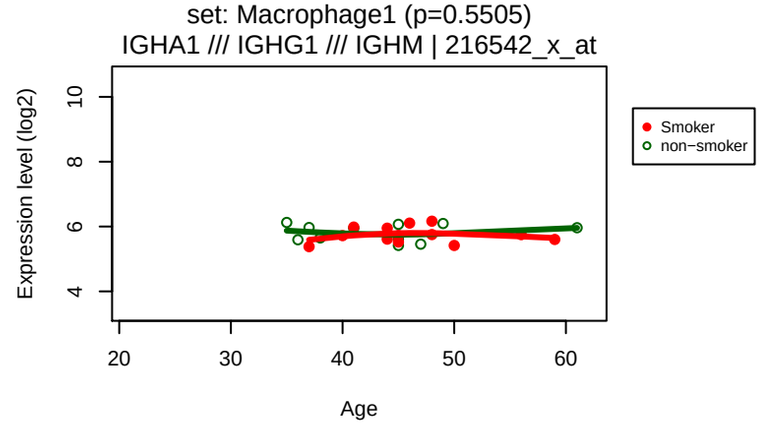

set: SmallAirway1 (p=0.2484)
IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Expression level (log2)
Age
Smoker
non−smoker

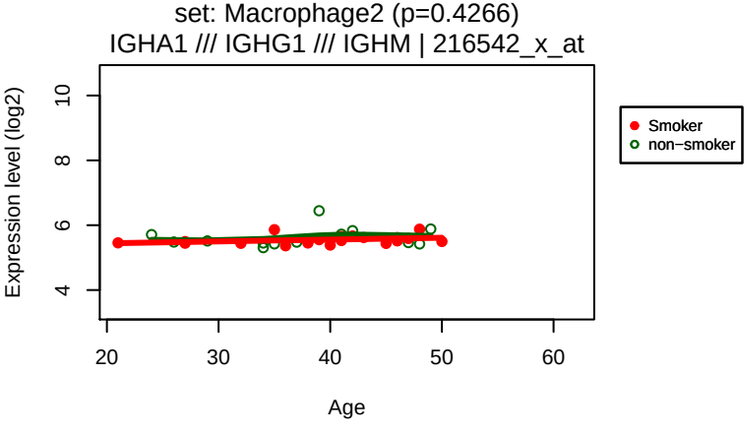

set: SmallAirway2 (p=0.9395)
IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Expression level (log2)
Age
Smoker
non−smoker

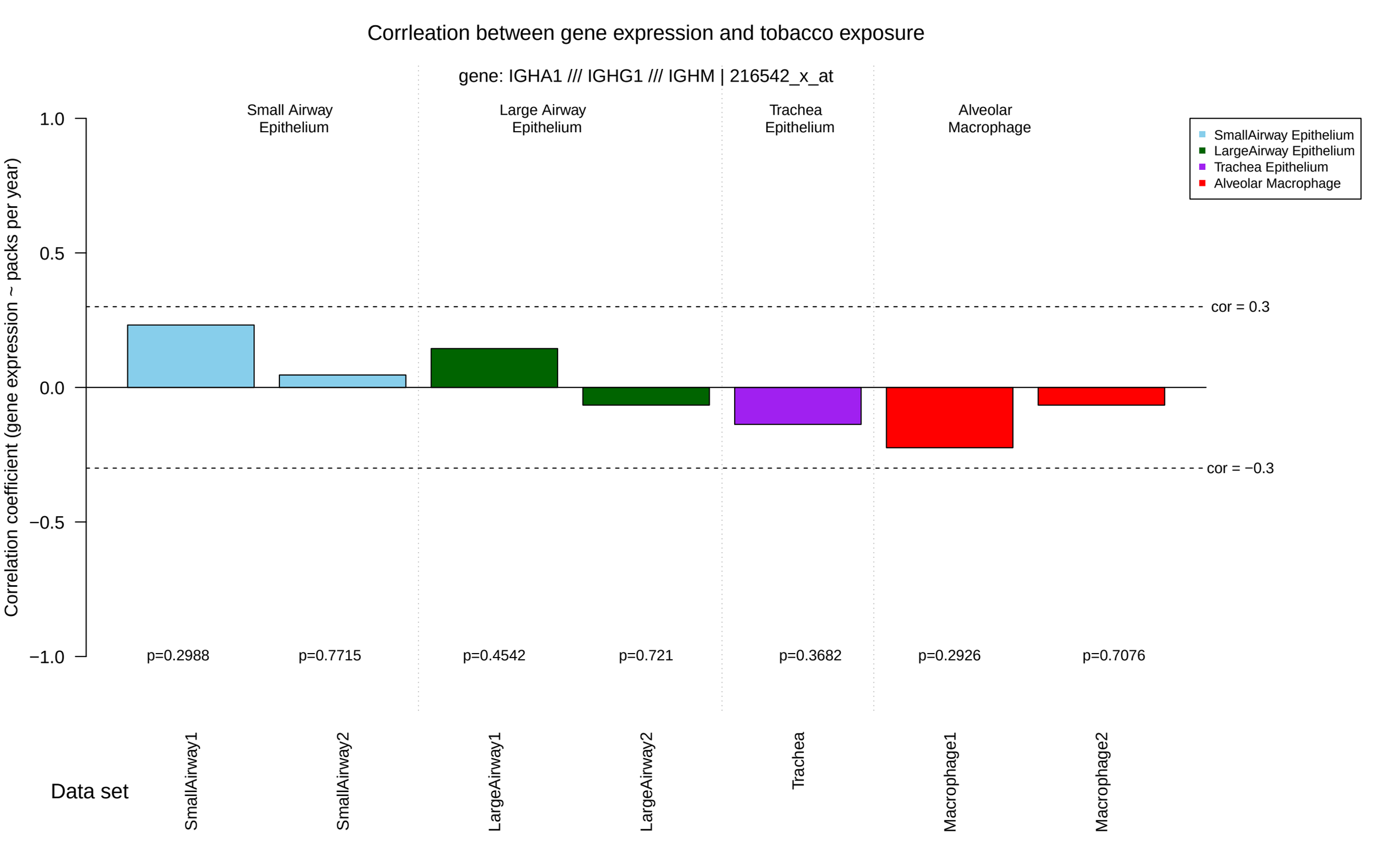

set: LargeAirway1 (p=0.0704)
IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8677)
IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4472)
IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5505)
IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4266)
IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHG1 /// IGHM | 216542_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
Correlation coefficient (gene expression ~ packs per year)
cor = 0.3
cor = −0.3
p=0.2988
p=0.7715
p=0.4542
p=0.721
p=0.3682
p=0.2926
p=0.7076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2